理论力学习题讲解（Xuanpu Yang）

理论力学杨望 2024

1. 约束在光滑平面$x+y+z=0$上的质量为m的质点，重力加速度沿$-g$方向

（1）自由度

（2）拉氏量

(1) $3-1=2$

(2) ① 选取广义坐标为 x, y 则 $z=-x-y$ ⇒ 建立其他坐标与广义坐标的联系

$T=\frac{1}{2}m\dot{x}^2+\frac{1}{2}m\dot{y}^2+\frac{1}{2}m\dot{z}^2$　$V=-mgz$　$L=\frac{1}{2}m\dot{x}^2+\frac{1}{2}m\dot{y}^2+\frac{1}{2}m(-\dot{x}-\dot{y})^2+mg(x+y)=m\dot{x}^2+m\dot{y}^2+m\dot{x}\dot{y}+mg(x+y)$

2. 耦合振子$L=\frac{1}{2}(\dot{x}^2+\dot{y}^2+\dot{z}^2)+a\dot{x}\dot{y}-\frac{1}{2}k(x^2+y^2+z^2)$

（1）导出运动方程

（2）求出 3 个简正频率

(1) $\frac{\partial L}{\partial \dot{x}}=\dot{x}+a\dot{y}$　$\frac{\partial L}{\partial \dot{y}}=\dot{y}+a\dot{x}$　$\frac{\partial L}{\partial \dot{z}}=\dot{z}$

$\frac{\partial L}{\partial x}=kx$　$\frac{\partial L}{\partial y}=ky$　$\frac{\partial L}{\partial z}=kz$

$$\begin{cases} \frac{d}{dt}\frac{\partial L}{\partial \dot{x}} = \frac{\partial L}{\partial x} \Rightarrow \ddot{x} + a\ddot{y} = kx \\ \frac{d}{dt}\frac{\partial L}{\partial \dot{y}} = \frac{\partial L}{\partial y} \Rightarrow \ddot{y} + a\ddot{x} = ky \\ \frac{d}{dt}\frac{\partial L}{\partial \dot{z}} = \frac{\partial L}{\partial z} \Rightarrow \ddot{z} = kz \end{cases}$$

⇒ 运动方程

法一：

$$\begin{cases} \ddot{x} + a\ddot{y} = kx \ ① \\ \ddot{y} + a\ddot{x} = ky \ ② \\ \ddot{z} = kz \end{cases} \Rightarrow \quad \Rightarrow \ \omega_1 = \sqrt{k}$$

(运动方程) 拼凑 ⇓ 核心在于凑出 $\ddot{X} + \omega^2 X = 0$

① + ② $\ddot{x} + \ddot{y} + a\ddot{x} + a\ddot{y} = k(x+y)$

$(1+a)(\ddot{x}+\ddot{y}) = k(x+y)$

令 $X = x+y$

$(1+a)\ddot{X} = kX$

$\omega_2 = \sqrt{\frac{k}{1+a}}$

① − ② $\ddot{x} - a\ddot{x} + a\ddot{y} - \ddot{y} = k(x-y)$

$(1-a)(\ddot{x}-\ddot{y}) = k(x-y)$

令 $Y = x-y$

$(1-a)\ddot{Y} = kY$

$\omega_3 = \sqrt{\frac{k}{1-a}}$

法二：(不找简正坐标)

$$L = \frac{1}{2}(\dot{x}^2+\dot{y}^2+\dot{z}^2) + a\dot{x}\dot{y} - \frac{1}{2}k(x^2+y^2+z^2)$$

$$= \frac{1}{2}(\dot{x}^2+\dot{y}^2+\dot{z}^2+2a\dot{x}\dot{y}) - \frac{1}{2}k(x^2+y^2+z^2)$$

$$T = \frac{1}{2}(\dot{x},\dot{y},\dot{z}) \underbrace{\begin{pmatrix} 1 & a & \\ a & 1 & \\ & & 1 \end{pmatrix}}_{M} \begin{pmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{pmatrix}$$

$$V = \frac{1}{2}(x,y,z) \underbrace{\begin{pmatrix} k & & \\ & k & \\ & & k \end{pmatrix}}_{K} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

⇒ 同时把 T 和 V 对角化

$\ddot{X} + \omega^2 X = 0 \Rightarrow \frac{1}{2}M\dot{X}^2 + \frac{1}{2}KX^2 \Rightarrow \omega = \sqrt{\frac{K}{M}}$

$$(K - \omega^2 M)\begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$$

$$\det|K - \omega^2 M| = 0 \Rightarrow \begin{vmatrix} k-\omega^2 & -a\omega^2 & 0 \\ -a\omega^2 & k-\omega^2 & 0 \\ 0 & 0 & k-\omega^2 \end{vmatrix}$$

久期行列式

令 $k-\omega^2 = s \Rightarrow s^3 - a^2\omega^4 s = 0 \Rightarrow s(s^2 - a^2\omega^4)$

解得 $s=0 \quad s=+a\omega^2 \quad s=-a\omega^2$

$s=0 \Rightarrow k-\omega_1^2=0 \Rightarrow \omega_1 = \sqrt{k}$

$s = a\omega^2 \Rightarrow k-\omega_2^2 = a\omega_2^2 \Rightarrow \omega_2 = \sqrt{\frac{k}{1+a}}$

$s = -a\omega^2 \Rightarrow k-\omega_3^2 = -a\omega_3^2 \Rightarrow \omega_3 = \sqrt{\frac{k}{1-a}}$

哈密顿力学部分： $L=\frac{m}{2}(u^2+v^2)(\dot{u}^2+\dot{v}^2)-\frac{1}{2}qE(u^2-v^2)$

（1）通过勒让德变换导出哈密顿量$H=\frac{p_v^2+p_u^2}{2m(v^2+u^2)}+\frac{1}{2}Eq(u^2-v^2)$

（2）导出正则方程

诺特定理部分

无穷小连续变换（ε为无穷小参数）：

$$u'=u+\frac{u\varepsilon}{u^2+v^2}$$

$$v'=v-\frac{v\varepsilon}{u^2+v^2}$$

（3）证明在这个变换下动能$T=\frac{1}{2}m(u^2+v^2)(\dot{u}^2+\dot{v}^2)$不变

（4）证明在这个变换下势能做如下变换：

$$V'=V+Eq\varepsilon$$

（5）由上述结论证明该变换为对称性变换（给了作用量的定义）

（6）由拉格朗日力学的诺特定理（给了拉格朗日力学中诺特定理守恒量的表达式和每一项的定义）证明$Q=\frac{p_u u-p_v v}{u^2+v^2}+Eqt$为守恒量

（7）验证Q的确能生成上述变换（仅需验证正则坐标）

（正则变换）

(1) $L = \frac{m}{2}(u^2+v^2)(\dot{u}^2+\dot{v}^2) - \frac{1}{2}qE(u^2-v^2)$ 勒让德 ⇒ $\dot{q}$ 换成 p

$$p_u = \frac{\partial L}{\partial \dot{u}} = m(u^2+v^2)\dot{u} \Rightarrow \dot{u} = \frac{p_u}{m(u^2+v^2)}$$

$$p_v = \frac{\partial L}{\partial \dot{v}} = m(u^2+v^2)\dot{v} \Rightarrow \dot{v} = \frac{p_v}{m(u^2+v^2)}$$

$$H = p_u\dot{u} + p_v\dot{v} - L = \frac{p_u^2}{m(u^2+v^2)} + \frac{p_v^2}{m(u^2+v^2)} - \frac{m}{2}(u^2+v^2)\left(\frac{p_u^2}{m^2(u^2+v^2)^2} + \frac{p_v^2}{m^2(u^2+v^2)^2}\right) + \frac{1}{2}qE(u^2-v^2)$$

(不可出现速度 只可出现动量)

$$= \frac{p_u^2}{2m(u^2+v^2)} + \frac{p_v^2}{2m(u^2+v^2)} + \frac{1}{2}qE(u^2-v^2)$$

(2)
$$\begin{cases} \dot{p}_u = -\frac{\partial H}{\partial u} \\ \dot{p}_v = -\frac{\partial H}{\partial v} \\ \dot{u} = \frac{\partial H}{\partial p_u} \\ \dot{v} = \frac{\partial H}{\partial p_v} \end{cases} \Rightarrow \begin{cases} \dot{u} = \frac{p_u}{m(u^2+v^2)} \\ \dot{v} = \frac{p_v}{m(u^2+v^2)} \end{cases}$$

$$\begin{cases} \dot{p}_u = +\frac{p_u\dot{u}+p_v\dot{v}}{2m(u^2+v^2)^2}\cdot 2u + qEu \\ \dot{p}_v = \frac{p_u\dot{u}+p_v\dot{v}}{2m(u^2+v^2)^2}\cdot 2v - qEv \end{cases}$$

$$\Rightarrow \begin{cases} \dot{p}_u = \frac{(p_u\dot{u}+p_v\dot{v})u}{m(u^2+v^2)^2} + qEu \\ \dot{p}_v = \frac{(p_u\dot{u}+p_v\dot{v})v}{m(u^2+v^2)^2} - qEv \end{cases}$$

(3) ☆☆☆ $T = \frac{1}{2}m(u^2+v^2)(\dot{u}^2+\dot{v}^2)$ $T' = \frac{1}{2}m(u'^2+v'^2)(\dot{u}'^2+\dot{v}'^2)$

☆☆☆ $O(\varepsilon^1)$ ⇒ 保持不变 ☆☆☆ $O(\varepsilon^2)$ 忽略

$$\begin{cases} u' = u + \frac{u\varepsilon}{u^2+v^2} \\ v' = v - \frac{v\varepsilon}{u^2+v^2} \end{cases}$$ u v 都是 t 的函数

$$\begin{cases} \dot{u}' = \dot{u} + \frac{\dot{u}\varepsilon(u^2+v^2) - 2u\dot{u}u\varepsilon}{(u^2+v^2)^2} = \dot{u} + \frac{(v^2-u^2)\dot{u}\varepsilon - 2v\dot{v}u\varepsilon}{(u^2+v^2)^2} \\ \dot{v}' = \dot{v} - \frac{\dot{v}\varepsilon(u^2+v^2) - 2v\dot{v}v\varepsilon}{(u^2+v^2)^2} = \dot{v} - \frac{(u^2-v^2)\dot{v}\varepsilon - 2u\dot{u}v\varepsilon}{(u^2+v^2)^2} \end{cases}$$

(2u²u̇ε 与 −2vv̇uε 对 v 也求导；−2uu̇vε 与 −2uu̇vε 对 u 也求导)

$$\begin{cases} \dot{u}'^2 = \dot{u}^2 + \frac{2\dot{u}^2\varepsilon(v^2-u^2)}{(u^2+v^2)^2} \pm \frac{-4v\dot{v}u\varepsilon\dot{u}}{u^2+v^2\text{相消}} + \frac{(v^2-u^2)^2\dot{u}^2\varepsilon^2}{(u^2+v^2)^2} & O(\varepsilon^1) \\ \dot{v}'^2 = \dot{v}^2 + \frac{2\dot{v}^2\varepsilon(v^2-u^2)}{(u^2+v^2)^2} + \frac{+4u\dot{u}v\varepsilon\dot{v}}{} + \frac{(v^2-u^2)\dot{v}^2\varepsilon^2}{(u^2+v^2)^2} & O(\varepsilon^2) \end{cases}$$

$$\begin{cases} u'^2 = u^2 + \dfrac{2u^2\varepsilon}{u^2+v^2} + \cancel{\dfrac{u^2\varepsilon^2}{(u^2+v^2)^2}} & O(\varepsilon^2) \\ v'^2 = v^2 - \dfrac{2v^2\varepsilon}{u^2+v^2} + \cancel{\dfrac{v^2\varepsilon^2}{(u^2+v^2)^2}} & O(\varepsilon^2) \end{cases}$$

$T' = \frac{m}{2}(u'^2+v'^2)(\dot{u}^2+\dot{v}^2)$ (原来的部分分离出来) ☆☆☆

$$= \frac{m}{2}\left(u^2+v^2 + \frac{2u^2\varepsilon}{u^2+v^2} - \frac{2v^2\varepsilon}{u^2+v^2}\right)\left[\dot{u}^2+\dot{v}^2 + \frac{2\dot{u}^2\varepsilon(v^2-u^2)}{(u^2+v^2)^2} + \frac{2\dot{v}^2\varepsilon(v^2-u^2)}{(u^2+v^2)^2}\right]$$

$$= \frac{m}{2}(u^2+v^2)(\dot{u}^2+\dot{v}^2) + \frac{m}{2}\left(\frac{2(u^2-v^2)\varepsilon}{u^2+v^2}\right)(\dot{u}^2+\dot{v}^2) + \cancel{\frac{2(u^2-v^2)\varepsilon}{(u^2+v^2)}\cdot\frac{2\varepsilon(v^2-u^2)(\dot{u}^2+\dot{v}^2)}{(u^2+v^2)^2}} - 2v\dot{v}u\varepsilon - 2u\dot{u}v\varepsilon$$

$$+ \frac{m}{2}(u^2+v^2)\frac{2\varepsilon(v^2-u^2)(\dot{u}^2+\dot{v}^2)}{(u^2+v^2)^2}$$

$\Downarrow$ $O(\varepsilon^2)$

$$= \frac{m}{2}(u^2+v^2)(\dot{u}^2+\dot{v}^2) + \cancel{\frac{m}{2}\cdot\frac{2\varepsilon(u^2-v^2)(\dot{u}^2+\dot{v}^2)}{(u^2+v^2)}} - \cancel{\frac{m}{2}\frac{2\varepsilon(u^2-v^2)(\dot{u}^2+\dot{v}^2)}{(u^2+v^2)}}$$

0

$$= \frac{m}{2}(u^2+v^2)(\dot{u}^2+\dot{v}^2) = T$$

(4) $V = \frac{1}{2}qE(u^2-v^2)$

$V' = \frac{1}{2}qE(u'^2-v'^2)$

$$= \frac{1}{2}qE\left(u^2 + \frac{2u^2\varepsilon}{u^2+v^2} - v^2 + \frac{2v^2\varepsilon}{u^2+v^2}\right)$$

$$= \frac{1}{2}qE(u^2-v^2) + \frac{1}{2}qE\cdot\frac{2\varepsilon(u^2+v^2)}{u^2+v^2}$$ (把原来部分分离出来)

$$= \frac{1}{2}qE(u^2-v^2) + qE\varepsilon$$

$$= V + qE\varepsilon$$

(5) 对称性变换

$S_2 - S_1 = F(q,t)$

S_2 ⇓ 变换后　S_1 ⇓ 变换前

$$S = \int_{t_i}^{t_f} L\,dt = \int_{t_i}^{t_f}(T-V)\,dt$$

① 验对称性变换，得 F

② 写出守恒荷

③ $\left.\frac{\partial q'}{\partial\varepsilon}\right|_{\varepsilon=0}$　$\left.\frac{\partial t'}{\partial\varepsilon}\right|_{\varepsilon=0}$

④ 代入

$$S_2=\int_{t_i}^{t_f}(T'-V')\,dt=\int_{t_i}^{t_f}(T-V-qE\varepsilon)\,dt$$

$$S_1=\int_{t_i}^{t_f}(T-V)\,dt \qquad L'=L+\frac{df(q,t)}{dt} \qquad S'=S+f(q,t)$$

$$S_2-S_1=\int_{t_i}^{t_f}-qE\varepsilon\,dt=-qE\varepsilon(t_f-t_i)$$

$$=\int_{t_i}^{t_f}\varepsilon\left[\frac{d}{dt}(-qEt)\right]$$

S在相差边界项 $F=-qEt$ 的意义下保持不变

因此为对称性变换

(6) $Q_A=\sum_i\frac{\partial L}{\partial\dot q_i}\left.\frac{\partial q'}{\partial\varepsilon_A}\right|_{\varepsilon_A=0}+\sum_i\left(L-\frac{\partial L}{\partial\dot q_i}\dot q_i\right)\left.\frac{\partial t'}{\partial\varepsilon_A}\right|_{\varepsilon_A=0}-F$ $\qquad O(\varepsilon^1)$ $\qquad O(\varepsilon^2)$

$\frac{\partial L}{\partial\dot q_i}\Rightarrow P$

$\left(L-\frac{\partial L}{\partial\dot q_i}\dot q_i\right)\Rightarrow -H$

$-F$ ⇒ 拉氏量不唯一性

时空性质

- 伽利略变换（洛仑兹变换）→ 观测者速度
- 电磁场 → 规范变换

$$\begin{cases}u'=u+\dfrac{u\varepsilon}{u^2+v^2}\\ v'=v-\dfrac{v\varepsilon}{u^2+v^2}\end{cases}$$

$$\left.\frac{\partial t'}{\partial\varepsilon}\right|_{\varepsilon=0}=0 \qquad \left.\frac{\partial u'}{\partial\varepsilon}\right|_{\varepsilon=0}=\frac{u}{u^2+v^2} \qquad \left.\frac{\partial v'}{\partial\varepsilon}\right|_{\varepsilon=0}=-\frac{v}{u^2+v^2} \qquad F=-qEt$$

$$p_u=\frac{\partial L}{\partial\dot u}=m(u^2+v^2)\dot u \qquad Q_A=m(u^2+v^2)\dot u\frac{u}{(u^2+v^2)}-m(u^2+v^2)\dot v\frac{v}{u^2+v^2}$$

$$p_v=\frac{\partial L}{\partial\dot v}=m(u^2+v^2)\dot v \qquad =m\dot u u-mv\dot v$$

$$\Rightarrow Q_A=\frac{p_u u-p_v v}{(u^2+v^2)}+qEt$$

(7) 守恒荷为生成元（正则变换）

$$\begin{cases}\dfrac{\partial u'}{\partial\varepsilon}=[u,Q]=\dfrac{\partial u}{\partial u}\dfrac{\partial Q}{\partial p_u}+\dfrac{\partial u}{\partial v}\dfrac{\partial Q}{\partial p_v}-\dfrac{\partial u}{\partial p_u}\dfrac{\partial Q}{\partial u}-\dfrac{\partial u}{\partial p_v}\dfrac{\partial Q}{\partial v}=\dfrac{\partial Q}{\partial p_u}=\dfrac{u}{(u^2+v^2)}\\ \dfrac{\partial v'}{\partial\varepsilon}=[v,Q]=\dfrac{\partial v}{\partial u}\dfrac{\partial Q}{\partial p_u}-\dfrac{\partial v}{\partial p_u}\dfrac{\partial Q}{\partial u}+\dfrac{\partial v}{\partial v}\dfrac{\partial Q}{\partial p_v}-\dfrac{\partial v}{\partial p_v}\dfrac{\partial Q}{\partial v}=\dfrac{\partial Q}{\partial p_v}=\dfrac{-v}{(u^2+v^2)}\end{cases}$$

$$\begin{cases} \frac{\partial u'}{\partial \varepsilon} = \frac{u}{(u^2+v^2)} \\ \frac{\partial v'}{\partial \varepsilon} = \frac{-v}{(u^2+v^2)} \end{cases} \Rightarrow \quad u' = \frac{u\varepsilon}{u^2+v^2} + u \qquad v' = \frac{-v\varepsilon}{u^2+v^2} + v$$

生成元

$$\begin{cases} \frac{\partial q'}{\partial \varepsilon} = [q, Q] \\ \frac{\partial p}{\partial \varepsilon} = [p, Q] \end{cases}$$

正则方程形式

$$\begin{cases} \frac{dq}{dt} = -\frac{\partial H}{\partial p} \\ \frac{dp}{dt} = \frac{\partial H}{\partial q} \end{cases} \Rightarrow \begin{cases} \frac{\partial q}{\partial \varepsilon} = -\frac{\partial Q}{\partial p} \\ \frac{\partial p}{\partial \varepsilon} = \frac{\partial Q}{\partial q} \end{cases}$$

也用G

4. 系统哈密顿量为$H = -p\sin 2q$

变换：

$$P = -2\sqrt{p}\sin q$$

$$Q = \sqrt{p}\cos q$$

（1）用泊松括号验证这个变换是正则变换

（2）证明变换后的哈密顿量为$\tilde{H}(Q,P)=PQ$

（3）写出$\tilde{H}$的哈密顿雅可比方程

（4）用分离变量法求解该方程

(1) $\begin{cases} P=-2\sqrt{p}\sin q \\ Q=\sqrt{p}\cos q \end{cases}$ $\quad H=-p\sin 2q$

$[q,p]=1 \quad [q,q]=0 \quad [p,p]=0$

$\Rightarrow [Q,Q]_{pq}=0 \quad [P,P]_{pq}=0 \Rightarrow \frac{\partial P}{\partial q}\frac{\partial P}{\partial p}-\frac{\partial P}{\partial p}\frac{\partial P}{\partial q}=0$（可不写）

只需验证 $[Q,P]_{pq}=1 \Rightarrow$ Q，P用旧变量表示

$$[\sqrt{p}\cos q,-2\sqrt{p}\sin q]_{pq}=\frac{\partial(\sqrt{p}\cos q)}{\partial q}\frac{\partial(-2\sqrt{p}\sin q)}{\partial p}-\frac{\partial(\sqrt{p}\cos q)}{\partial p}\frac{\partial(-2\sqrt{p}\sin q)}{\partial q}$$

$$=-\sqrt{p}\sin q\left(-\frac{2}{2\sqrt{p}}\sin q\right)-\frac{\cos q}{2\sqrt{p}}(-2\sqrt{p}\cos q)$$

$$=\sin^2 q+\cos^2 q$$

$$=1$$

在新旧变量表述下 $[Q,P]_{pq}=1 \Rightarrow$ 是正则变换

(2) $H=-p\sin 2q$ $\quad PQ=\sqrt{p}\cos q\,(-2\sqrt{p}\sin q)=-p\sin 2q$

$\tilde{H}=PQ$

(3) $H(Q,\frac{\partial S}{\partial Q},t)+\frac{\partial S}{\partial t}=0$

其中 $P=\frac{\partial S}{\partial Q}$ $\quad H=Q\frac{\partial S}{\partial Q}$

HJE: $Q\frac{\partial S}{\partial Q}+\frac{\partial S}{\partial t}=0$ $\quad$ (解 S

(4) 由于 $\tilde{H}$ 不含时 $\Rightarrow$ E 守恒

$$\begin{cases}\tilde{H}=Q\frac{\partial S}{\partial Q}=E\\ \frac{\partial S}{\partial t}=-E\ \checkmark\end{cases}$$

$\Rightarrow S=-Et+W(Q)+A$

$\frac{\partial S}{\partial Q}=\frac{\partial W}{\partial Q}$

$\tilde{H}=Q\frac{\partial W}{\partial Q}$

$Q\frac{\partial W}{\partial Q}=E \Rightarrow \frac{\partial W}{\partial Q}=\frac{E}{Q} \Rightarrow W=E\ln Q$

$S=-Et+E\ln Q+A$ (A 为常数)